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Abstract

This report presents an deep and careful comparison between two important link analysis algorithms: PageRank and HITS. Both methods were implemented and tested on multiple datasets, including a social network graph, a Facebook friendship network, and a Wikipedia link graph. The objective was to evaluate how each algorithm identifies influential nodes in complex networks and to examine their respective computational characteristics, convergence behaviors, and limitations.

Experiments showed that PageRank effectively measures global node importance through random walks and converges efficiently, but it tends to favor highly connected or older nodes. On the other hand HITS distinguishes between two complementary roles (hubs and authorities) capturing local relevance within directed networks but being sensitive to graph structure and slower convergence.

Additional experiments explored Personalized, Weighted, and Hybrid PageRank variants, which improved contextual ranking and recommendation potential in user centric networks. Overall, the presented report highlights how PageRank provides stable, scalable global rankings, while HITS offers finer structural insights at the expense of higher computational cost and sensitivity to topology.

Introduction

The scope of this report is to present our observations about 2 algorithms related to Link Analysis. We will compare PageRank and HITS, highlighting the strengths, weaknesses, and applications of each. After testing them on different datasets, we will observe their performance, implementation and possible improvements.

Datasets Overview

0.1 Social Network Epinions dataset

The first dataset used is an undirected social network graph consisting of Facebook nodes. The format shows which node is connected to which. Each node corresponds to a user, and each directed edge indicates that one user trusts another. It contains $\approx$ 70,000 nodes and 508,837 directed edges The network structure captures asymmetric relationships, meaning trust is not necessarily mutual

0.2 Wikipedia dataset

The Wikipedia dataset used in this implementation consists of 70,273 pages, each treated as an individual node within a large interconnected network. Hyperlinks between pages define the edges, forming a directed graph where the connectivity of each node directly influences its importance in the network. During preprocessing, each page was parsed to extract its title and all outbound links, creating a structured representation that includes the source page, its out-degree, and a list of destination pages. Pages with no outbound links were removed to ensure meaningful contributions to the link graph.

From a structural perspective, the dataset exhibits characteristics typical of a complex, scale-free network. Certain pages act as hubs with many connections, while the majority of nodes have relatively few links. This heterogeneity naturally biases random-walk algorithms like PageRank toward high-degree nodes, reflecting the underlying network topology. The connectivity patterns also suggest modularity, where clusters of related topics are interconnected, making the network both dense in local regions and sparse globally.

0.3 Facebook dataset

This dataset is very similar to the social network one presented in the beginning. The only difference is the source, this dataset containing specific friendship relations among individuals on Facebook.

PageRank

PageRank represents arguably the most popular link analysis algorithm. The core idea is that nodes are assigned a greater score if many other nodes in the network link to them.

Algorithm

```
import networkx as nx
# Load the graph
G = nx.read_edgelist("../datasets/facebook_combined.txt",
create_using=nx.Graph(), nodetype=int)
G = G.to_directed()
# Compute PageRank
pagerank_scores = nx.pagerank(G, alpha=0.85)
# Best 10 nodes by score
top_nodes = sorted(pagerank_scores.items(), key=lambda x: x[1], reverse=True)[:10]
for node, score in top_nodes:
    print(f"Node {node}: PageRank = {score:.5f}")
```

0.4 Results & Observations

0.4.1 Social Network Dataset

Node	PageRank Score
18	0.00466
737	0.00288
1719	0.00215
790	0.00212
118	0.00206
136	0.00204
143	0.00200
40	0.00159
1619	0.00153
4415	0.00148

Table 1: Top 10 nodes from social network dataset

This algorithm will work by performing a random walk through the network. However, even though the graph is undirected, there is a specific level of subjectivity given by the number of connections for each node, thus the probability of landing on a specific node directly depends on the number of links that node has. This way, high degree nodes will get high degree score, as we can notice from the results above.

One strength we can talk about in the first place is robustness: even if the graph is undirected, the results are meaningful, showing the influential entities in the network.

Another interesting strength is global relevance, meaning that the algorithm is not tied to any specific query; and stable convergence, proven by the fact that PageRank converges nicely for graphs of large size as well.

As longs as weaknesses are concerned, one problem arises. Nodes that accumulate many incoming links (or that are old in the network) tend to get higher PageRank, a "rich get richer" effect. This way it is harder for new nodes to compete with them.

Simplicity is an important factor in this algorithm, which however leaves room for another weakness: the lack of interpretability. Essentially, all links are treated equally, even in the case where a specific one can be more relevant than another, which can be a major drawback in social contexts.

0.4.2 Wikipedia Algorithm

$$PR(u) = \frac{1-d}{N} + d \sum_{v \in B_u} \frac{PR(v)}{L(v)}$$

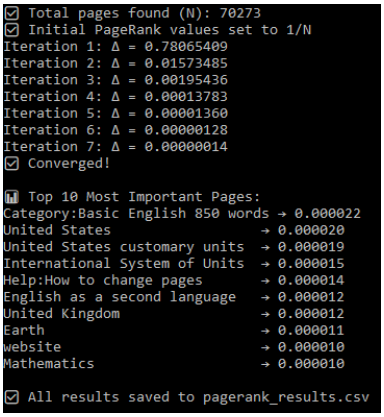


Figure 1: Code output

The PageRank algorithm successfully processed a total of 70,273 pages from the dataset, treating each page as a distinct node within a large interconnected web of links. Initially, every page was assigned an equal PageRank value of 1/N (= 0.00001423), representing an equal probability of being visited. Over successive iterations, the algorithm redistributed these ranks based on the structure of links between pages. The decreasing delta values — from 0.78065409 in the first iteration down to 0.00000014 by the seventh iteration — indicate that the system rapidly stabilized, ultimately converging after just seven iterations. This shows that the PageRank values reached equilibrium, meaning each page's importance became consistent relative to others.

Page	PageRank Score
Category:Basic English 850 words	0.000022
United States	0.000020
United States customary units	0.000019
International System of Units	0.000015
Help:How to change pages	0.000014
English as a second language	0.000012
United Kingdom	0.000012
Earth	0.000011
website	0.000010
Mathematics	0.000010

Table 2: Top 10 Wikipedia pages ranked by PageRank score.

The final results highlight the 10 most influential or well-connected pages in the dataset. At the top is "Category:Basic English 850 words" (0.000022), followed by "United States" (0.000020), "United States customary units" (0.000019), and "International System of Units" (0.000015), among others such as "Help:How to change pages" and "English as a second language." These pages appear frequently across other articles, acting as key hubs within the network — either due to their broad relevance (e.g., "United States," "Earth," "Mathematics") or structural significance within the dataset (e.g., "Help:How to change pages"). Although the PageRank values themselves are small, typically between 0.000010 and 0.000022, this is expected since the total probability mass of 1.0 is distributed across more than seventy thousand nodes. Consequently, even small numerical differences represent meaningful variations in relative importance. Overall, the results demonstrate that the algorithm performed correctly — converging efficiently and identifying the pages that serve as the most central and influential within the network.

0.5 Parameter setting

0.5.1 Damping Factor α

- Within PageRank, we avoid $\alpha = 1$. Otherwise, this would entail a user that exclusively follows only links. Not only is this unrealistic, but more importantly this increases the chances of reaching dead ends and spider traps.
- On the other spectrum, $\alpha = 0$ is also not recommended. This value would mean the user ignores all links. Which is an unrealistic setting given the context of the dataset. Moreover, this uniformity would diminish the importance of certain links.
- The value of $\alpha = 0.85$ as chosen by Google closely reflects human behavior and strikes a good balance between "Link following" vs "teleportation".

This value also allows for relatively more efficient convergence of the power iteration whilst maintaining valuable link-based relations

0.6 Key factors that affect performance

This report measures PageRank's performance in terms of 2 factors: Computational speed and result quality (whether the results are appropriate and stable). The following are several key factors affecting PageRanks's speed and quality.

0.6.1 Graph Size/Type

- The bigger the input graph (number of nodes and links), the larger the transition matrix used to compute the convergence. The relative size of the input graph therefore positively correlates with more memory needed and longer computational duration.
- Whether the graph is sparse or dense is another factor. Sparse graphs result in faster computational duration as most matrix entries would be 0.

0.6.2 Damping factor α

- The value of α affects number of iteration till convergence and hence computational speed.
- Having a higher α value can result in convergence being reached relatively slower, requiring more iteration.
- A lower convergence can result in a faster convergence duration, however promotes more uniformity, which may not provide valuable insight to which nodes are most connected.

0.6.3 Convergence tolerance

- Convergence tolerance directly affects the number of iterations required to reach convergence.
- Lower convergence tolerance would reflect more accurate PageRank Values for each node, but at the cost of more iterations. Resulting in longer computational durations. Conversely, higher tolerance values will allow convergence to be reached relatively faster.

0.7 Failure cases

Observation: When an undirected graph was given, PageRank failed to give nodes with distinct/meaningful rankings. The algorithm also threw errors occasionally, but more concerning was the discrepancy between node rankings when given directed vs. undirected graph. The following are possible reasons for this failure case and what they entail.

0.7.1 Link Symmetry

- Within an undirected graph, all nodes will have their "n-links" equal their "out-links", making each connection bidirectional. This opposes the concept of a "random surfer" model as there now lacks a distinction between "being linked from" vs "linking to". This holds significant difference (and is what defines PageRank).

0.7.2 Loss of Teleportation

- No dead-ends exist within an undirected graph as all edges are bi-directional. Although this does not affect the mathematical validity of the power iteration convergence and weight distribution, this does remove the value of teleportation and the need to decide the "Alpha" α value. In summary, an undirected graph contradicts a fundamental building block the PageRank algorithm is designed for under - "teleportation".

0.8 Advanced

In social media platforms such as Facebook, the graph structure is inherently **user-centric**. Each user serves as the center of their own network, interacting with friends, followers, and groups. Consequently, PageRank variations that incorporate *consumer behavior*, *personalization*, and *interaction dynamics* are best suited to improve recommendation quality, content relevance, and influence detection.

This section explores advanced PageRank algorithms—namely **Personalized PageRank**, **Weighted PageRank**, and a combined hybrid approach—to better capture the behavioral and relational patterns observed in user-centric networks.

0.9 Personalized PageRank

The **Personalized PageRank** algorithm extends the standard PageRank model by focusing on a targeted subset of nodes rather than the entire network. Instead of uniformly random restarts, the algorithm restarts more frequently from a user's personalized subset—such as friends, frequent contacts, or members of a shared group.

This modification allows the computed PageRank values to reflect a user's specific interests, behaviors, and relationship patterns. The resulting rankings emphasize nodes that are most relevant to the individual, making this approach particularly effective for personalized recommendations, friend suggestions, and content prioritization within social media environments.

Let a social network graph be defined as:

$$G = (V, E)$$

where V is the set of nodes and E is the set of edges.

Let:

- A be the adjacency matrix (unweighted by default),
- D be the diagonal out-degree matrix of A, with entries

$$D_{ii} = \sum_j A_{ij},$$

- d be the damping factor (typically $d = 0.85$),
- p be the personalized restart vector of size $|V|$,
- r be the Personalized PageRank vector.

The iterative form of Personalized PageRank is:

$$r = d \cdot A^\top D^{-1} r + (1 - d) \cdot p$$

where the personalized restart vector p is defined as:

$$p_i = \begin{cases} \dfrac{1}{|S|}, & \text{if node } i \in S, \\ 0, & \text{otherwise,} \end{cases}$$

with $S \subseteq V$ being the user-specific personalized set of nodes.

Explanation: This formulation is similar to standard PageRank, but the random jump is biased toward the personalized set S rather than being uniform across all nodes.

0.10 Weighted PageRank

In real-world social networks, not all connections hold equal importance. Some users engage more actively through frequent likes, comments, messages, or shared interactions. The **Weighted PageRank** algorithm accounts for this by assigning different weights to edges according to interaction frequency, duration, or recency.

Edges representing more frequent or meaningful interactions are given higher weights, contributing more significantly to the PageRank computation. Consequently, nodes (users) that maintain stronger or more consistent engagement receive higher scores. This approach captures the depth of social ties and interaction intensity, producing rankings that more accurately reflect real-world social influence and activity.

In a weighted network, not all edges are equal. Let:

- w_{ij} be the weight of the edge from node i to node j (e.g., interaction frequency),
- W be the weighted adjacency matrix,

- D_w be the diagonal matrix of weighted out-degrees, with entries

$$(D_w)_{ii} = \sum_j w_{ij},$$

- d be the damping factor (typically $d = 0.85$),
- r be the Weighted PageRank vector,
- $\mathbf{1}$ be the uniform teleport vector (can also be personalized if desired).

Node-wise Formulation

For each node $i \in V$:

$$r_i = (1-d) + d \sum_{j \in N(i)} \frac{w_{ji}}{\sum_{k \in N(j)} w_{jk}} r_j$$

Matrix Formulation

$$r = d \cdot W^\top D_w^{-1} r + (1-d) \cdot \mathbf{1}$$

Explanation: Stronger edges contribute more to a node's rank. Frequent interactions increase the influence of a node in the network.

0.11 Hybrid Personalized and Weighted PageRank

The **Hybrid Personalized and Weighted PageRank** combines the benefits of personalization and interaction weighting to produce a more realistic model of social connectivity. This approach restricts the analysis to a single user's local network while assigning weights to edges based on interaction frequency.

By integrating these two concepts, the hybrid model effectively balances user-specific preferences with the strength of social interactions. It highlights users who are both personally relevant and actively engaged, thereby emphasizing meaningful, high-frequency relationships over passive or weak connections. This hybrid approach provides a computationally efficient yet socially accurate method for modeling engagement and recommending connections or content.

Personalized Weighted PageRank (Hybrid)

We can combine both edge weights and personalization in a single PageRank formulation. Let:

- W be the weighted adjacency matrix, where w_{ij} captures the intensity of interaction from node i to node j,

- D_w be the diagonal matrix of weighted out-degrees, with entries

$$(D_w)_{ii} = \sum_j w_{ij},$$

- d be the damping factor (typically $d = 0.85$),
- p be the personalized restart vector (biased toward a user-specific set $S \subseteq V$),
- r be the resulting hybrid Personalized Weighted PageRank vector.

Matrix Formulation

$$r = d \cdot W^\top D_w^{-1} r + (1 - d) \cdot p$$

Explanation: This hybrid formulation favors nodes that are both strongly connected (high-weight edges) and relevant to the user (nodes in the personalized set S).

0.12 Reference Data Set

- **Second Data Set:** srinathsai/Google-pagerank-algorithm-on-Wikipedia — a memory-efficient implementation of the PageRank algorithm used to identify important nodes or pages in large-scale networks such as Wikipedia.
- This repository serves as a reference for studying advanced PageRank algorithms and evaluating their performance on real-world datasets.

0.13 Evaluation Perspective

When applied to large graph structures such as social networks or the Wikipedia dataset, these advanced PageRank methods aim to:

- Identify highly influential or relevant nodes within the network.
- Capture both user personalization and interaction intensity in ranking outcomes.
- Improve recommendation quality and contextual relevance.
- Maintain scalability and efficiency for large, real-world datasets.

By leveraging both personalization and weighting, the resulting PageRank scores provide a deeper understanding of social connectivity and influence—enabling more accurate and user-aware recommendation systems.

HITS

HITS is a strong algorithm used for link analysis. In this procedure, every node has 2 main roles: authority and hub. Certain scores are appointed for each of the 2. A node with a high authority score tends to be a good source of information, while a good hub score is awarded when the node points to many good authorities.

Algorithm

```
import networkx as nx
# Load graph
G = nx.read_edgelist("../datasets/social_graph.txt", create_using=nx.DiGraph(), nodetype=int
G = G.to_directed()
# Compute HITS scores
hubs, authorities = nx.hits(G, max_iter=100, normalized=True)
# Top 10 hubs
top_hubs = sorted(hubs.items(), key=lambda x: x[1], reverse=True)[:10]
for node, score in top_hubs:
    print(f"Node {node}: Hub Score = {score:.5f}")
# Top ten authorities
top_auths = sorted(authorities.items(), key=lambda x: x[1], reverse=True)[:10]
for node, score in top_auths:
    print(f"Node {node}: Authority Score = {score:.5f}")
```

0.14 Results & Observations

Node	Hub Score
645	0.00252
634	0.00189
44	0.00146
27	0.00127
563	0.00123
637	0.00119
824	0.00117
763	0.00116
71399	0.00115
1059	0.00114

Table 3: Top 10 HITS Hub Score from social network dataset

Node	Authority Score
18	0.00445
143	0.00249
1179	0.00242
34	0.00239
401	0.00234
737	0.00230
550	0.00214
136	0.00210
27	0.00209
1	0.00207

Table 4: Top 10 HITS Authority Score from social network dataset

After running these experiments, some conclusions regarding the performance of HITS can be drawn. AS we can see, node 18 has the highest authority

Node	Hub Score
Wikipedia:List of common misspellings	0.994881
Wikipedia:Basic English ordered wordlist	0.047039
Wikipedia:Basic English alphabetical wordlist	0.047004
Wikipedia:List of 1000 basic words	0.042022
2006	0.014841
2004	0.014462
2007	0.013960
1999	0.012134
January 25	0.011251
March 28	0.010789

Table 5: Top 10 HITS Hub Score from the Wikipedia dataset

Node	Authority Score
country	0.019222
government	0.018780
year	0.017915
money	0.017907
weather	0.017687
business	0.017460
language	0.017438
town	0.017429
military	0.017251
rule	0.017244

Table 6: Top 10 HITS Authority Score from the Wikipedia dataset

score, which can also translate as the 'highest PageRank score'. By looking at the nodes, we can clearly see that hubs are differentiated from authorities:

Hubs: 645, 634, 44, 27 ...

Authorities: 18, 143, 1179, 34 ...

This way, nodes that only point to important entities are differentiated from relevant authorities. For example ode 645 is a top hub but does not appear in the top authorities, showing that HITS captures a role-based distinction that PageRank can not.

Another area where HITS proves its strength is given by the fact that it highlights locally important nodes in terms of information flow: hubs point to authorities, authorities are pointed to by hubs. This is useful in networks like social networks with directed edges or citation networks. Node 18 tops the authority score, which also correlates with high PageRank.

Nonetheless, by looking at the hub list, nodes that act as information aggregators or recommenders can be easily identified. By looking at the authority list, we can see the most "trusted" or influential nodes. This is a specific view that single-score algorithms like PageRank can nott provide.

As long as weaknesses are concerned, firstly we can consider the sensitivity to link patterns.HITS only propagates importance along directed links. In this way, if a node points to many low quality nodes,then its hub score may be inflated if the authority scores are not well distributed. Conversely, a node that should be an authority but is poorly linked by hubs may have a lower score than expected.

Secondly we can consider the issue of localized perspective. HITS emphasizes mutual reinforcement between hubs and authorities, which can make it focus on dense subgraphs. In this manner, less connected authorities may be underrated even if they are globally influential. For example, some nodes with decent PageRank do not appear in the top hub/authority lists because they are not part of dense hub-authority clusters.

Another problem to mention is that HITS is computationally iterative and query-sensitive and it requires iteration to convergence. In large graphs, convergence can be slower than PageRank, especially if the graph has many disconnected components. While we computed it on the full graph, in practice, HITS is usually query dependent: it works best on a subgraph relevant to a topic. Here we applied it globally, which works but loses some of the intended focus.

Finally, HITS can be sensitive to spam or noisy edges, because in networks with fake or artificial links, hubs can point to many "low-quality" nodes, boosting their hub score. Similarly, authorities can be artificially cited by low-quality hubs. PageRank's damping factor helps mitigate this, but unfortunatelly HITS has no equivalent mechanism.

0.14.1 Wikipedia Dataset Analysis

As per Tables 5 and 6 above, applying the HITS algorithm to a subset of Wikipedia revealed a clear distinction between navigational hub pages and authoritative content pages. The page "Wikipedia:List of common misspellings" achieved a dominant hub score (0.9949), highlighting its role as a major navigational entry point linking to numerous authoritative pages. Other high-scoring hubs, like "Wikipedia:Basic English ordered wordlist" and "Wikipedia:List of 1000 basic words," similarly serve as list-oriented references, while year-based pages (e.g., 2004, 2006, 1999) provide minor cross-linking.

Authority scores were led by general concept pages such as country, government, money, and weather. Their relatively uniform scores (0.017–0.019) indicate influence is distributed across semantically central topics rather than concentrated in a single node, reflecting a well-connected subgraph.

The results illustrate HITS's dual structure: hubs link to multiple authorities, and authorities gain prominence through links from reliable hubs. Overall, the algorithm effectively distinguishes between navigational indices (hubs) and key informational content (authorities), capturing the hierarchical and semantic organization of Wikipedia's hyperlink network.

0.15 Parameter setting

0.15.1 Convergence tolerance

- Similar to PageRank, Convergence tolerance is also an important parameter setting within HITS.
- A smaller tolerance can increase the time taken to reach convergence, but provide more minute and precise scores for both hub and authority. Conversely, a higher tolerance value reaches convergence faster but is less accurate.

0.15.2 "Normalized"

- Setting this parameter to true prevents the HITS algorithm from updating Hub and Authority values without bound.
- Normalization rescales the transposed/updated Hub and authority scores. This prevents the values from becoming too large in magnitude and allows convergence to a unique value instead.
- more importantly this results in scores that are **relative**and not absolute values.

0.16 Key factors that affect performance

0.16.1 "Graph size"

- Similar to Page Rank, a larger graph size would lead to slower computational speeds due to more costly matrix operations. Additional, unlive PageRank, HITS takes into account 2 eigen vector computation.

0.16.2 "Convergence Tolerance"

- Similar to PageRank, convergence tolerance is a tradeoff between precision/accuracy of hub/authority score and iterations. With a smaller tolerance leading to a more precise result but more iterations and vice versa for a larger tolerance value.

0.16.3 "Normalization"

- Normalization affects HITS in various ways other than just computational speed. Without normalization, matrix values of hub/authority scores are allowed to increase in magnitude without bound. Normalization ensures a stable convergence towards a value to curb this issue.

0.17 Failure cases

0.17.1 "Uniformity in Undirected Graphs"

- When given an undirected graph, hub/authority scores often converge into identical values. Which provides no distinction of the 2 separate roles for nodes.
- This is attributed to how HITS mathematically calculates scores for it's nodes. Within undirected graphs, every edge becomes bi-directional and mutual. Mathematically, both hub and authority will now produce the same matrix when multiplied. This defeats the purpose of the HITS algorithm in separating Hub vs Authority.
- Due to symmetry in values, distinction in scores for hub and authority is absent.

0.17.2 "Topic Drift"

- Since HITS usually operates on a fixed, root set of pages instead of global search, HITS can be observed to offer results that diverges from the original topic point.
- This is attributed to HITS algorithms pulling in hubs that structurally sound (robust number of links) but not meaningfully relevant. This allows for a mutual reinforcement loop that goes deeper into unrelated topics, eventually leading to topic drift

0.18 Advanced

The original Hyperlink-Induced Topic Search (HITS) algorithm computes two scores for each node i in a directed graph:

$$a_i = \sum_{j:(j \to i)} h_j, \qquad h_i = \sum_{j:(i \to j)} a_j$$

where a_i and h_i are the authority and hub scores respectively. Several extensions of HITS have been proposed to improve its robustness, personalization, and resistance to spam. This section introduces three notable variants: **Stochastic HITS**, **HITS with Priors**, and the **TrustRank–HITS Hybrid**.

0.18.1 Stochastic HITS

Motivation: The original HITS is deterministic and can suffer from instability and topic drift. To mitigate this, Stochastic HITS introduces a random jump mechanism, similar to the teleportation concept in PageRank.

Modified Equations:

$$a^{(t+1)} = \alpha A^T h^{(t)} + (1-\alpha)a^{(0)}, \qquad h^{(t+1)} = \alpha A a^{(t)} + (1-\alpha)h^{(0)}$$

Where:

- A is the adjacency matrix of the graph.
- $\alpha \in [0, 1]$ is the damping factor controlling the random jump probability.
- $a^{(0)}$ and $h^{(0)}$ are prior (initial) authority and hub distributions.

Interpretation: With probability α, the algorithm follows standard link-based updates, and with probability $(1 - \alpha)$, it "restarts" from the prior distribution. This makes HITS more stable, smoother, and less sensitive to noise.

0.18.2 HITS with Priors

Motivation: Standard HITS treats all nodes equally during initialization. HITS with Priors integrates topic or user-specific information through a prior probability vector.

Modified Equations:

$$a^{(t+1)} = A^T h^{(t)} + \lambda p_a, \qquad h^{(t+1)} = A a^{(t)} + \lambda p_h$$

Where:

- p_a and p_h represent prior distributions for authorities and hubs.
- λ controls the balance between learned scores and prior information.

Interpretation: Nodes with higher prior probabilities receive additional importance, leading to topic-sensitive or personalized rankings. This variant is analogous to Personalized PageRank but for both hub and authority propagation.

0.18.3 TrustRank–HITS Hybrid

Motivation: HITS can be exploited by link spam or malicious nodes. By integrating trust information from TrustRank, the algorithm can downweight untrusted nodes.

Modified Equations:

$$a_i = \sum_{j:(j \to i)} t_j \cdot h_j, \qquad h_i = \sum_{j:(i \to j)} t_j \cdot a_j$$

Matrix Form:

$$a^{(t+1)} = A^T (T \odot h^{(t)}), \qquad h^{(t+1)} = A (T \odot a^{(t)})$$

where T is a diagonal matrix containing trust scores t_i, and $\odot$ denotes element-wise multiplication.

Interpretation: Links from trusted nodes contribute more to authority and hub updates, whereas links from low-trust nodes contribute less. This hybrid method effectively reduces the influence of spam or unreliable nodes.

Conclusion

Through the comparative analysis of PageRank and HITS, it was observed that both algorithms can identify influential nodes in an effective manner even in large networks, but with distinct strengths. PageRank gives a global measure of node importance based on link popularity and performs well across diverse graph structures, proving stable convergence and robustness to noise. However, its uniform treatment of links can oversimplify relationships and favor older or highly connected nodes.

HITS, on the other hand, offers a 2-way perspective by distinguishing hubs from authorities, which makes it more suitable for networks where directional information flow is critical. However, HITS is more sensitive to graph topology, convergence tolerance, and the presence of dense subgraphs, which can lead to localized bias or topic drift.

Extensions such as Personalized, Weighted, and Hybrid PageRank demonstrate how link analysis can be adapted for social or recommendation-based contexts by incorporating user preferences. In conclusion, while PageRank remains superior for global ranking and scalability, HITS provides valuable insight into role-based relationships within the network.